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REDOX TITRATION LAB WORKSHEET

Show all work (except average and standard deviation). Turn in this worksheet before leaving the lab. Reminder: never round values prematurely; only round to the correct number of significant figures when *reporting final answers* and use *unrounded values* for subsequent calculations.

Prepare an Excel spreadsheet to tabulate and compute the answers to the following questions; for each question, you will need to show all work for **just 1 trial** - remember that doing the first calculation by hand will make it easier to write the spreadsheet formula for the remaining trials.

1. **Data analysis – Part 1**
   (a) Show one, full sample calculation for the determination of concentration of the $KMnO_4$ solution based on your titration against FAS.

   $[KMnO_4](M) =$

   (b) Collect at least 6 total trials (including yours) for the $KMnO_4$ concentration. Report the average and standard deviation here, and calculate and report the relative standard deviation.

   $\overline{[KMnO_4]}(M) =$ RSD =

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(c) The listed tolerance for a 50 mL burette is ± 0.03 mL, which means that delivering 35 mL from a burette has a $\pm 0.09\%$. Analytical balances have an uncertainty of ± 0.0001 g, which means that a 0.1 g sample has $\pm 0.1\%$ relative uncertainty.

Do either of these sources account for the total uncertainty as estimated by your RSD above? If not, suggest another reasonable source of error that could account for the experimental uncertainty. Explain in no more than 1 - 2 sentences total.

2. **Data analysis – Part 2**

(a) Using your titration data from Part 2, calculate the actual (observed) molar concentration of the *diluted* peroxide. Show all work for 1 trial.

$[H_2O_2]_{\text{dilute}}(M) =$

(b) For the same trial as part (a) above, calculate the original, *undiluted* molar concentration of peroxide according to your data.

$[H_2O_2]_{\text{original}}(M) =$

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(c) Collect at least 6 total trials (including yours) for the original H_2O_2 concentration. Report the average and standard deviation here, and calculate and report the relative standard deviation.

$\overline{[H_2O_2]}_{\text{original}}(\text{M}) =$	RSD =

(d) Convert the manufacturer's value for the concentration of the store-bought peroxide into molarity. You can assume that peroxide is almost entirely water and has a density of $1.00\,\frac{\text{g}}{\text{mL}}$.

$[H_2O_2]_{\text{manufacturer}}(\text{M}) =$

(e) Calculate the percent relative difference between your calculated average molarity and the manufacturer's molarity for the peroxide solution. Explain your choice of percent relative difference formula in no more than 1 sentence.

%diff =

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(f) Based on the similarity/difference between your results and the manufacturer's value, what can you conclude about the hydrogen peroxide solution? Explain in no more than 1 sentence.

3. **Questions for thought**

(a) Calculate the actual volume (in mL) of 4.5 M sulfuric acid required in a titration using 0.7500 g of FAS. Based on this, why is it not necessary to be careful when measuring the sulfuric acid for these titrations? Explain in no more than 1 sentence.

$V(H_2SO_4)(\text{mL}) =$

(b) For titrations using a 50-mL burette, it is most appropriate for the titrant volumes used to be between 35 and 45 mL. Smaller volumes are acceptable, though less than ideal. Volumes greater than 50 mL are not appropriate. Explain these practices using what you know about uncertainty in measurements in no more than 1 - 2 sentences.